

Muhammed Muminul Hoque

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House no. 549/I, Moushumi R/A, DC Road, West Bakalia, Chattogram, Bangladesh

Education

Bachelor of Science in Computer Science & Engineering

University of Dhaka, Bangladesh

February 2019 – March 2024

GPA: 3.35/4.00
(3.58 in last 60 credits)

Research Experience

Research Intern

November 2025 – Present

Under the supervision of *Professor Yanfu Zhang*

Department of Computer Science, College of William & Mary, Williamsburg, VA

- Developing a "Position-Aware" In-Context Learning (ICL) evaluation pipeline for long-context LLMs to mitigate performance degradation ("Context Rot") in multi-shot prompting.
- Co-authored *DyV2X*, a dynamic graph learning framework for V2X cooperative trajectory prediction, targeting ICRA 2027. (**2nd author**).
- Designed a GRU history encoder across 6 CTDG architectures (TGN, TGAT, GraphMixer, DyGFormer, DyRep, JODIE) on the V2X-Seq-TFD dataset; achieved minADE = 0.952 m, minFDE = 1.664 m, MR = 24.7%.

Publications

Published

E. Hossain, P. R. Saha, **M. M. Hoque**, M. S. Islam, J. Al Faruk, and M. A. Hossain, **ECCE 2025**
“*Privacy Preserving and Secure Digital Identity Verification Mechanism: AES-Encrypted Smart NID with Two-Factor Protection Against Unauthorized Access,*”
in *Proc. 4th Int. Conf. on Electrical, Computer and Communication Engineering (ECCE)*, CUET, Chattogram, Bangladesh, 2025, pp. 1–6.
doi:10.1109/ECCE64574.2025.11012996

P. R. Saha, Md. S. Islam, T. K. Saha, **M. M. Hoque**, A. Haque, E. Hossain, **STI 2025**
“*Quantum-Resistant Blockchain: Enhancing Blockchain Security with ROUND5, A Lattice-based Cryptosystem,*”
in *Proc. IEEE 7th Int. Conf. on Sustainable Technologies for Industry 5.0 (STI 2025)*.
doi:10.1109/STI69347.2025.11367567

- Implemented ROUND5 lattice-based cryptography to reduce post-quantum vulnerability; proposed an efficient key exchange protocol suitable for SMEs.

Accepted

Jonayed Al Faruk, **Muhammed Muminul Hoque**, Md. Nawaz Shorif, Susmoy Paul, Md. Bodrul Islam, Joyonto Das, **BIM 2026**
“*Toward Accurate Urban Temperature Forecasting in Seoul: A Hybrid Ensemble of Bias-Corrected Machine and Deep Learning Models with Explainable AI,*”
in *Proc. 3rd Int. Conf. on Big Data, IoT and Machine Learning (BIM 2025)*, Springer Lecture Notes in Networks and Systems, Vol. 1, 2026, accepted for publication.

- Explored XGBoost, KNN, CNN, SVR, and Feedforward Neural Networks; applied SHAP and LIME for explainability of meteorological and topographic features.

Muhammed Muminul Hoque, Md. Masudur Rahman, Jonayed Al Faruk, A.K.M. Bahalul Haque, Mir Tasrif Ahmed, MD. Nayeem Pervez **ECCT 2026**
“A Comparative Analysis of Traditional and Ensemble Machine Learning Models for Phishing Email Detection with Explainable AI.”
 in *Proc. Int. Conf. on Electrical, Computer and Communication Technologies (ECCT 2026)*, Dhaka International University, Bangladesh, May 7–9, 2026. To be published in a Taylor & Francis Book.

- Built a unified dataset of 290,404 labeled emails from 12 public corpora.
- Benchmarked classical vs. ensemble ML models with explainable AI (XAI); achieved ROC-AUC of 94.6%.

Submitted

Yiran Ding, **Muhammed Muminul Hoque**, Maiqi Jiang, Yongming Qin, Sidi Lu, Yanfu Zhang, **ICRA 2027**
(Under Review)
“DyV2X: Dynamic Interaction Graphs for V2X Cooperative Trajectory Prediction,”
 submitted to *IEEE Int. Conf. on Robotics and Automation (ICRA 2027)*, Seoul, Republic of Korea, May 24–28, 2027.

- Proposed a dynamic interaction graph framework that captures the temporal evolution of inter-agent interactions for V2X cooperative motion forecasting.
- Demonstrated consistent outperformance over prior methods, with particularly significant gains in safety-critical turning scenarios.

Research in Progress

Muhammed Muminul Hoque, Md. Masudur Rahman, A.K.M. Bahalul Haque **(In Prep)**
“ZEAL: A Zero-Shot Ensemble Framework for Adaptive Phishing Detection with LLMs”
Preparing to Submit (Elsevier Journal).

- Developed a multi-tier agentic pipeline utilizing heterogeneous model ensembles and reasoning escalation for zero-shot detection — zero training data required.
- Achieves 98.44% accuracy on 4,812 samples across two benchmarks, outperforming fine-tuned transformer baselines through advanced logical consensus.

Muhammed Muminul Hoque, Maiqi Jiang, Y. Zhang, et al., **“Optimizing In-Context Learning via Non-Linear Attentional Priors in Long-Context Windows.”** **(In Prep)**
Target: Top-tier AI/NLP Venues (e.g., ACL/NAACL 2027 or ICLR 2027).

- Proposed an inference-only method using empirical attentional analysis to optimize example positioning in long-context ICL, significantly recovering performance lost in mid-context.
- Benchmarked on state-of-the-art LLMs across 4k+ token context lengths, demonstrating consistent accuracy gains over naive relevance-based ordering.

Project Work

Aurora: Autonomous Academic Research Agent optimized for low-resource servers **2026**

- Engineered a multi-agent AI system optimized for heavily constrained edge hardware (1GB RAM), utilizing Open-Router to dynamically route tasks between specialized LLMs (Hermes 405B, Qwen, Gemini Flash) with zero local memory overhead.
- Built an asynchronous Telegram interface featuring real-time token streaming, computer vision (document parsing), and automated synthesis of ArXiv research papers.
- Implemented a robust background task scheduler integrating multi-backend deadline tracking across local JSON, Notion APIs, and Google Calendar.
- Tools: Python, AsyncIO, python-telegram-bot, APScheduler, Gemini API.

PeerTranslate: Verified Research Paper Translation for Every Language **2026**

- Open-source AI translation tool for research papers with a 4-pass back-translation verification pipeline (Translate

→ Back-Translate → Score → Re-translate), producing per-section confidence scores.

- Powered by Google Gemini with community-curated domain glossaries (ML, CS, general academic) to prevent mistranslation of technical jargon; supports multiple languages with open contributions.
- Tools: Python, FastAPI, Google Gemini API.
- Live demo: peertranslate.onrender.com

Toxicity Analysis of 4chan's /pol/ Board Using OpenAI Moderation and Google Perspective APIs

2025

- Developed an end-to-end pipeline to detect and analyze toxic speech patterns using OpenAI Moderation API and Google Perspective API for multi-dimensional toxicity scoring.
- Conducted ethical data collection, quantitative analysis, and visualization to identify trends in online discourse.
- Tools: Python, Pandas, Matplotlib.
- Reviewed by Prof. Kai-Cheng Yang (Binghamton University).

Machine Learning Based Speech Emotion Recognizer

2024

- Identifies and classifies emotions from speech, providing real-time emotional insights.
- Tools: JupyterLab, Python.
- Supervisor: Prof. Md. Alamgir Hossain, Professor (CSE) & Principal.

Technical Skills

Languages & Frameworks: Python, C/C++, Bash, PyTorch

Generative AI & LLM Systems: In-Context Learning (ICL), Multi-Agent Systems, vLLM, OpenRouter, Gemini API, Claude API, Prompt Engineering, Zero-Shot Inference

Graph Neural Networks & Motion Forecasting: TGN, TGAT, GraphMixer, DyGFormer, GCN, GAT, PyG, DGL, HiVT, DenseTNT, V2X-Graph, WOMD evaluation

ML & Explainability: Scikit-learn, SHAP, LIME, GNNExplainer, XGBoost, Random Forest, SVM

Tools & Libraries: Pandas, NumPy, Matplotlib, Git, GitHub, Overleaf, Google Colab

Training Experiences & Certifications

Cyber Security For Professionals, Byte Capsule

Feb 2023 – Aug 2023

- Six-month program covering threat analysis, risk management, network security, cryptography, and ethical hacking through hands-on exercises and expert-led lectures.

Standardized Test Scores

International English Language Testing System (IELTS)

14 December 2024

- Overall: 6.5 Listening: 8.0 Reading: 6.0 Writing: 6.0 Speaking: 6.0

Extra-curricular Activities, Awards, and Achievements

- MEC Computer Club (January 2023 – March 2024)
- 3rd Place at MEC CSE Carnival 2020 Programming Competition
- Batch Topper – 6-Month Self-learning Cybersecurity Training Program (Feb – Aug 2023)
- Champion of Invisible CTF 2023 (18 September 2023)
- Ranked in the Top 6% on TryHackMe

References

Prof. Yanfu Zhang

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